POWER-PAUSE(.)

-PRESENTED BY

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PROBLEM STATEMENT:

- In classrooms and hostels, lights and fans are often left ON even no one is present.
- This leads to unnecessary electricity wastage.
- It increases electricity bills and energy consumption.
- Manual switching depends on human responsibility.

NEED :

- Electricity is a limited and valuable resource.
- Many institutions face high electricity bill due to wastage.
- Manual control is not always reliable.
- An automated system can improve efficiency and reduce energy loss.

DESIGNED THINKING PROCESS :

1.Empathize:

Observed electricity wastage in classrooms and hostels.

2.Define:

Identified the need for an automated electricity control system.

3.Ideate:

Proposed AI-based human presence detection using webcam.

4.Prototype:

Developed a face detection system using python and openCV.

5. Test:

Tested the system by detecting room occupancy in real-time.

SOLUTION :

- The system detects human face using a laptop webcam.
- Face detection is implemented using openCV.
- If a face is detected, the system displays “Light ON”.
- If no face is detected,it displays “Light OFF”.
- This helps in reducing unnecessary electricity consumption.

TECHNOLOGY USED :

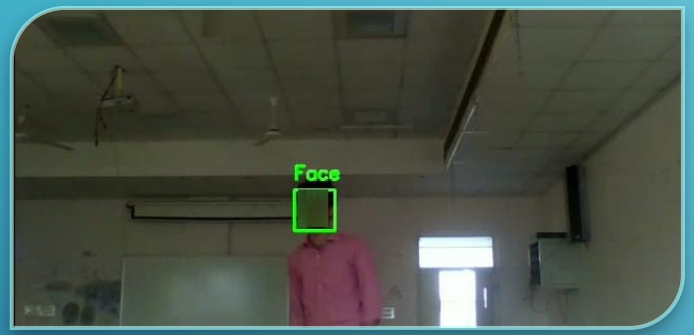
Python - Programming language used to develop the system.

- OpenCV – Library used for face detection and image processing.
- Haar Cascade Classifier – Pre-trained model used to detect frontal faces.
- Laptop Webcam – Captures real-time video input.

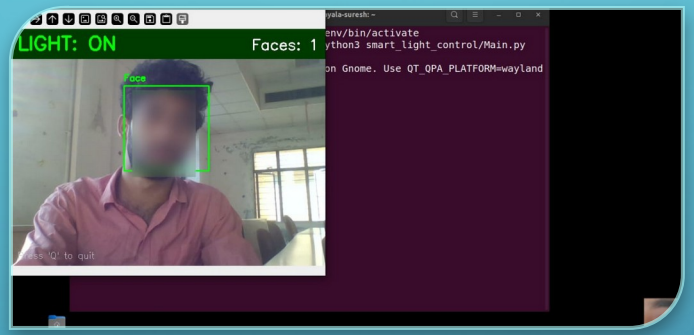
WORKING PRINCIPLE :

- The webcam continuously captures real-time video.
- Each video frame is processed using openCV.
- The haar cascade model detects human faces.
- If a face is detected then the system displays “Room Occupied-Light ON”.
- If no face is detected, then System displays “Room Empty-Light OFF”.

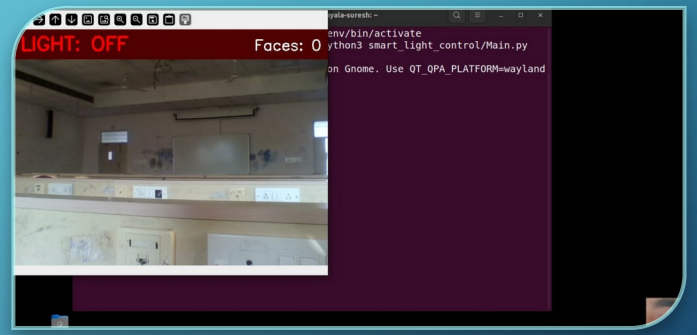
SAMPLE OUTPUT :



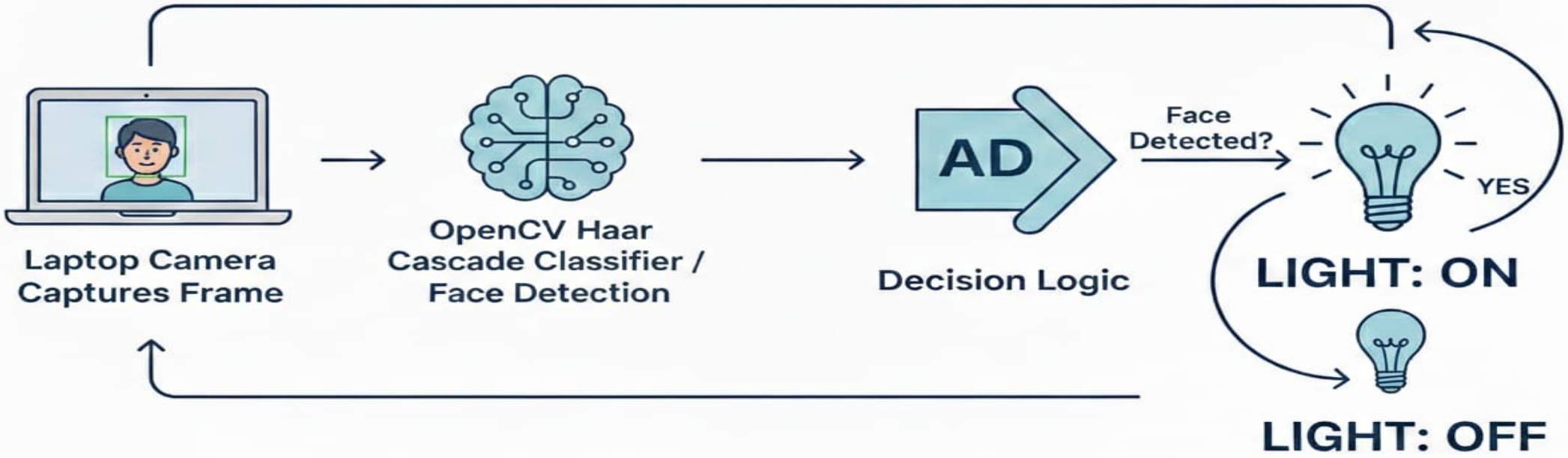
FACE DETECTED



LIGHT ON MODE



LIGHT OFF MODE



Real-time Face Detection for Automated Lighting

FUTURE ENHANCEMENTS :

- Improve detection accuracy for side faces.
- Make the system work in low lighting conditions.
- Connect the system to real classroom lights.
- Add motion sensors for better detection.
- Develop a mobile app for monitoring.

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GUIDING LIGHT : KONDAREDDY SIR

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Thanking you….